TBO Inventory Intelligence

An AI decision-support platform that tells TBO what travel inventory to pre-buy, how much, when, and at what price, to maximise risk-adjusted profit within a given budget.

87.5%

weekly demand forecast accuracy

+61%

realised profit vs last year's mix (back-test on simulated data)

< 1 s

to re-plan any budget, live in the browser

The problem: pre-buying travel inventory is a crore-rupee bet under uncertainty

27%

Sold-outs hide demand

of days are sold out. On those days bookings understate real demand by 36%, so naive plans under-buy exactly where it matters most.

585

Every rupee competes

options (45 hotels and flight routes × 13 travel weeks) compete for one fixed budget, each with its own margin, demand and risk.

15–60%

Unsold stock is a loss

is all TBO gets back on unsold allotments, and 4–16% of bookings cancel, so uncertainty has to be priced in.

TBO's question: with a given maximum budget, what should we pre-buy, how much, when, and at what selling price?

Figures from the prototype's dataset (Jan 2024 – Nov 2026, 10 destinations)

Our solution: one budget in, a ranked buy plan out

TBO sets

- Maximum budget (₹ Cr)
- Risk appetite: profit ↔ protect downside
- Max share per destination
- Hotels, flights or both
- Destinations to consider

The platform returns, for every position

What	hotel or flight route × travel week
How many	units to pre-buy (risk-adjusted quantity)
When	buy-by date at the cheapest supplier rate
Price	selling price vs the market
Profit	expected and worst-case profit
Why	demand drivers in plain English

Re-plans live in under a second · same answer as a professional MILP solver to within 0.2%

How it works: five steps from raw data to a buy plan

01

Recover true demand

On sold-out days, searches ÷ look-to-book ratio reveal what customers really wanted.

02

Forecast with uncertainty

P10 / P50 / P90 demand for every hotel and route, for each of the next 13 weeks.

03

Price, cost and cancellations

Weekly market price, price sensitivity, cancellation rates and the cheapest moment to buy.

04

Simulate 1,000 futures

Correlated demand scenarios turn every option into a full profit distribution.

05

Allocate the budget

An optimizer picks quantities to maximise profit while protecting the worst case.

Inputs: bookings · searches (look-to-book) · sold-outs · supplier quotes by lead time · cancellations · holidays and destination events

Stack: Python · LightGBM · PuLP + HiGHS · FastAPI · React

AI models and techniques

STEP	TECHNIQUE	WHAT IT GIVES TBO
Demand unconstraining	Look-to-book recovery on sold-out days	True demand, not capped bookings
Demand forecast	LightGBM quantile ensemble (daily + weekly)	P10 / P50 / P90 demand per item-week
Market price	LightGBM with holiday and event features	Weekly price including peak premiums
Price elasticity	Log-log regression on all days	Optimal selling price
Cancellations	LightGBM rate model	Net sellable demand
Supplier rates	LightGBM on quotes vs lead time	Buy-by date and unit cost
Risk	Monte Carlo, 1,000 correlated scenarios	Profit distribution and worst case
Allocation	Newsvendor + MILP (HiGHS), mean–CVaR	What and how many, within budget
Explainability	SHAP drivers + Claude API	Why each position is in the plan

Alignment with the project objectives

OBJECTIVE IN THE BRIEF	HOW WE DELIVER IT	EVIDENCE
Predict demand	Quantile forecasts from booking trends, search data, holidays and events; sold-out demand recovered	**87.5% weekly accuracy (naive: 81.3%)**
Identify opportunities	Opportunity score: ROI × sold-out pressure × confidence; destination × week heatmap	**532 profitable item-weeks ranked**
Optimize investments	MILP allocates the given budget across hotels, flights, destinations and weeks	**+28.5% expected profit vs last year's mix**
Maximize profitability	Optimal quantity, buy-by date and selling price for every position	**₹7.6 Cr saved by buying at the right time**
Manage risk	1,000 simulated futures, worst-case (CVaR) objective, risk slider	**Back-test: 3 of 495 positions lost money**

Expected outcome, delivered: a decision-support platform that takes a maximum budget as input and returns the right inventory, at the right time, within the right budget, for the best risk-adjusted profit.

Is it accurate? Every model scored on data it never saw

87.5%

weekly demand forecast accuracy

vs 81.3% for "same week last year". The best possible on this data is 89.8%, because day-to-day demand is partly random.

Time-based split: trained on 37,755 item-days, tested on 4,095 unseen item-days (11 Aug – 9 Nov 2026). No leakage: only data TBO would really observe.

MODEL	OURS	BASELINE
Sold-out demand recovery error on sold-out days	**9.0%**	36.3%
Supplier rates (buy timing) cost error	**1.8%**	6.5%
Market selling price weekly price error	**2.6%**	3.3%
Price elasticity mean absolute error	**0.154**	0.308
Forecast ranges P10–P90 coverage, target 80%	**81%**	74%

Back-test: would the plan really have made money?

₹11.74 Cr

realised profit on ₹64.1 Cr, within 3% of the ₹12.04 Cr predicted

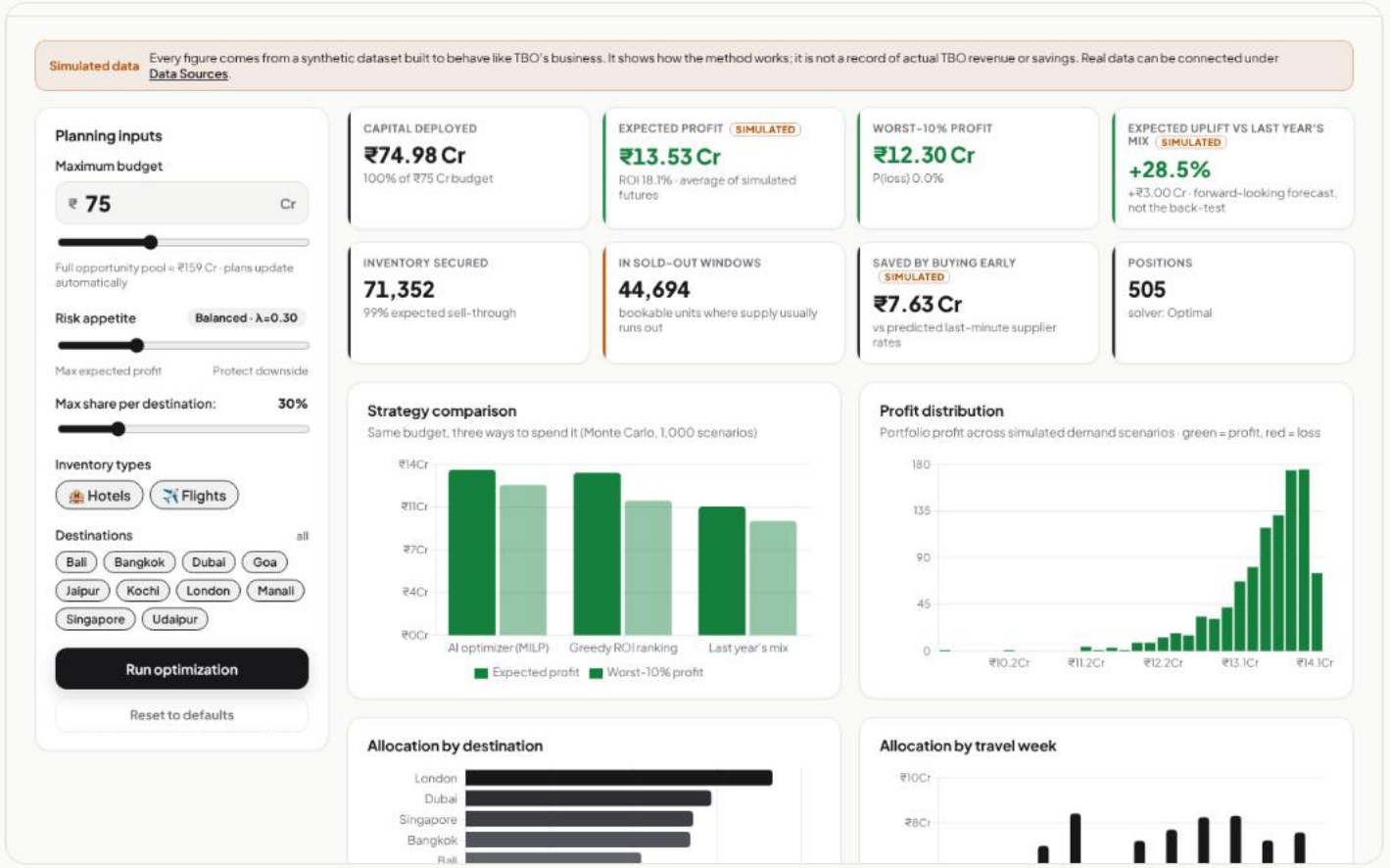
+61%

vs copying last year's mix

SAME ₹75 CR BUDGET	INVESTED	REALISED PROFIT	ROI	LOSING POSITIONS
AI optimizer (ours)	₹64.1 Cr	**₹11.74 Cr**	18.3%	**3**
Greedy ROI ranking	₹66.7 Cr	**₹11.98 Cr**	18.0%	**4**
Last year's mix	₹68.8 Cr	**₹7.30 Cr**	10.6%	**95**

Plan made on 11 Aug 2026 with only the data available that day, replayed against real demand, cancellations and supplier quotes for 12 travel weeks · synthetic dataset, so these are simulated results, not actual TBO profit

The prototype: a live decision-support dashboard



Live re-planning
change budget, risk or destinations and the plan updates in under a second

Explains every decision
demand drivers, events and a plain-English reason per position

Risk and budget strategy
risk–return frontier and diminishing-returns curve

Forecast explorer
history, recovered demand and 13-week forecast per item

Ready to use
copy or download the plan as CSV; light and dark mode; keyboard accessible

Live demo: tbo-inventory-optimizer-sggo.vercel.app · Code: github.com/srujanmh/tbo-inventory-optimizer

Impact for TBO, and what's next

+28.5%

expected profit vs last year's mix on a ₹75 Cr budget (simulated)

₹7.6 Cr

saved by buying at the cheapest lead time

44,000+

rooms and seats secured where supply usually sells out

99%

expected sell-through on pre-bought inventory

Next steps

1. Load TBO's real booking, search and supplier data through the data contract (validator built in) and retrain
2. Connect live feeds to the automated re-planner: searches, competitor rates, supplier availability
3. Use searches for future travel dates as an early demand signal
4. Connect supplier APIs for one-click allotment purchase

Demo: tbo-inventory-optimizer-sggo.vercel.app · Code: github.com/srujanmh/tbo-inventory-optimizer · Report and silent video: see README · Results are simulated